RUI ZHOU

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RESEARCH PROFILE

Materials Science PhD candidate (expected May 2027) working at the intersection of materials informatics, scientific machine learning, atomistic simulation, and experiment-linked data analysis. Develops active-learning and Bayesian-optimization workflows, predictive models, and reproducible scientific software for composition search, thin-film characterization, and multiscale materials R&D. Industry experience spans data-storage materials and lithium-ion battery development. Coauthor of 13 peer-reviewed publications, inventor on four granted patents and creator of four software applications.

EDUCATION

IOWA STATE UNIVERSITY — Ames, IA
PhD, Materials Science and Engineering — Expected May 2027

WASHINGTON UNIVERSITY IN ST. LOUIS — St. Louis, MO
M.S., Materials Science and Engineering — December 2019

CHANGCHUN UNIVERSITY OF SCIENCE AND TECHNOLOGY — Jilin, China
B.S., Materials Science and Engineering — June 2018

RESEARCH AND PROFESSIONAL EXPERIENCE

IOWA STATE UNIVERSITY — Ames, Iowa
Graduate Research Assistant — January 2023 - Present

- Use DFT and Machine Learning Potential molecular dynamics simulations to study atomic structures, interfaces, phase transitions, and ion transport in glassy and crystalline Li- and Na-based solid electrolytes, including thiophosphates, closo-hydroborates, and oxyhalides.
- Apply Bayesian optimization to glassy-electrolyte composition searches and couple machine-learning potentials with charge-equilibration models for interface studies; explore graph neural networks and diffusion models for glassy-material structure generation.
- Train machine-learning potentials using active-learning workflow with reference data from multiple electronic-structure methods, including constrained DFT for excited-state semiconductor modeling and MP2 for combustion chemistry; contribute to studies of ZnS/CdTe photoplasticity, ceramics, and metals.
- Build Python-based scientific-ML and Slurm/HPC workflows with NumPy, PyTorch, and MLX to manage large atomistic datasets and automate job preparation, execution, monitoring, post-processing, and model validation; prototype agentic orchestration for simulation campaigns.

SEAGATE TECHNOLOGY Normandale, Minnesota

Advanced Data Storage Material Discovery Intern May 2026 – Aug 2026

- Developed an end-to-end machine learning workflow for thin-film ellipsometry data analysis and materials characterization.
- Built graph-neural-network and RDKit-derived descriptor models for property prediction.
- Applied molecular dynamics simulation to study atomic structure and behavior in reader-head coating materials

Sunwoda Electric Vehicle Battery Co., ltd Shenzhen, China

Simulation Engineer December 2020 - December 2022

- Applied physics-based multiscale modeling across atomic, particle, electrode, and cell scales to support lithium-ion cathode development, electrode manufacturing, and cell-engineering decisions.
- Performed high-throughput DFT and force-field screening of dopants for NCM- and LFMP-based cathodes, translating atomistic predictions into candidate prioritization and materials-development decisions.
- Developed discrete-element models of cathode calendaring to evaluate how particle-size distributions and mixing ratios affect particle packing and electrode compaction, using physics-based optimization to guide process design.
- Built automated parameter-identification workflows across DEM, P2D/ECM electrochemistry, and FEM models, accelerating model calibration and simulation-feedback cycles.
- Developed predictive models using scikit-learn and symbolic regression for battery-power mapping and swelling-force prediction, providing computationally efficient estimates for cell-design evaluation.

PUBLICATIONS

K. Luo, **R. Zhou**, B. Li, and Q. An. *Facilitation of pyramidal dislocation slip in Mg–Bi alloys: A molecular dynamics study via machine learning potential.* Physical Review Materials 10(6), 063604 (2026). doi:10.1103/l83s-dwqz

R. Zhou, K. Luo, and Q. An. *Atomic-scale mechanisms of interphase formation at lithium–glassy sulfide electrolyte interfaces.* Journal of Materials Chemistry A 14, 22642–22654 (2026). doi:10.1039/D6TA01253A

R. Zhou, K. Luo, and Q. An. *Anion topology and cation–anion coupling govern phase transitions and ionic transport in closo-hydroborate solid electrolytes.* ACS Applied Energy Materials 9(8), 5148–5157 (2026). doi:10.1021/acsaem.6c00367

R. Zhou, K. Luo, and Q. An. *Data-driven atomistic modeling of crystalline and glassy solid-state electrolytes.* Chemical Communications 61(89), 17254–17270 (2025). doi:10.1039/D5CC04921K

K. Luo, **R. Zhou**, and Q. An. *Mechanistic insights into photoplasticity of CdTe.* International Journal of Mechanical Sciences 305, 110765 (2025). doi:10.1016/j.ijmecsci.2025.110765

J. Zeng, …, **R. Zhou**, …, L. Zhang, and H. Wang. *dpdata: A scalable Python toolkit for atomistic machine learning data sets.* Journal of Chemical Information and Modeling 65(21), 11497–11504 (2025). doi:10.1021/acs.jcim.5c01767

R. Zhou, K. Luo, L. Fei, and Q. An. *Unraveling superionic conductivity in Na2B12H12: Molecular dynamics study of phase transition, anion reorientation, and ionic conductivity via machine learning force field.* ACS Electrochemistry 1(2), 143–152 (2025). doi:10.1021/acselectrochem.4c00077

K. Luo, **R. Zhou**, and Q. An. *Unraveling photoplasticity in ZnS: Enhanced Peierls stress under photoexcitation using machine learning potentials.* ACS Materials Letters 7(1), 46–51 (2025). doi:10.1021/acsmaterialslett.4c01982

K. Luo, **R. Zhou**, S. W. Martin, and Q. An. *Flexible doorway-controlled Na+ ion diffusion in NaPSO glassy electrolytes from machine-learning force-field simulations.* Journal of Materials Chemistry A 12(48), 33518–33525 (2024). doi:10.1039/D4TA05071A

R. Zhou, K. Luo, S. W. Martin, and Q. An. *Insights into lithium sulfide glass electrolyte structures and ionic conductivity via machine learning force field simulations.* ACS Applied Materials & Interfaces 16(15), 18874–18887 (2024). doi:10.1021/acsami.4c00618

J. Liu, K. Luo, **R. Zhou**, and Q. An. *Understanding the role of 1/2 ⟨110⟩ dislocations in deformation mechanisms of single-crystal high-entropy carbide ceramics from machine learning force field simulations.* ACS Applied Engineering Materials 2(7), 1857–1865 (2024). doi:10.1021/acsaenm.4c00280

H.-G. Wang, C. Yuan, **R. Zhou**, Q. Duan, and Y. Li. *Self-sacrifice template formation of nitrogen-doped porous carbon microtubes toward high-performance anode materials in lithium-ion batteries.* Chemical Engineering Journal 316, 1004–1010 (2017). doi:10.1016/j.cej.2017.02.059

J. Guo, J. Liu, H. Dai, **R. Zhou**, T. Wang, C. Zhang, S. Ding, and H.-G. Wang. *Nitrogen-doped carbon nanofiber derived from polypyrrole-functionalized polyacrylonitrile for applications in lithium-ion batteries and the oxygen reduction reaction.* Journal of Colloid and Interface Science 507, 154–161 (2017). doi:10.1016/j.jcis.2017.07.117

PATENTS

Ternary material and preparation method thereof, lithium-ion battery, and power-utilization equipment.
Chinese invention patent CN114744187B; filed June 9, 2022; granted October 28, 2022.

Method, system, equipment, and storage medium for optimizing the particle size of material particles.
Chinese invention patent CN114169233B; filed November 30, 2021; granted January 7, 2025.

Simulation-optimization design method, system, equipment, and medium for battery-material mixing proportions.
Chinese invention patent CN115579085B; filed October 14, 2022; granted November 25, 2025.

Parameter-identification method, system, equipment, and medium for a secondary-battery physical model.
Chinese invention patent CN115392123B; filed August 26, 2022; granted August 8, 2025.

AWARDS

McGee-Wagner Materials Science and Engineering Scholarship 2025–2026